

About Me

Physicist turned AI engineer and researcher. My research is on mechanistic interpretability, mostly how concepts like refusal are represented geometrically inside language models. I also build applied AI systems, currently in synthetic market research and earlier in audio.

Experience

- Sep 2025–Present **AI Engineer**, *Synthetic Consumer Lab*
Sole engineer responsible for the full product stack: backend, frontend, AI systems, and statistical methodology. Architected the platform on a Laravel, Python/FastAPI, and Redis stack. Built RAG pipelines and synthetic consumer persona systems grounded in real demographic and behavioral data, alongside willingness-to-pay experiments and focus group simulations. Ensured AI outputs remain statistically valid and representative for market research applications.
- Sep 2025–Present **Teaching Assistant**, *Sabancı University*
Teaching assistant for Quantum Programming course, guiding students through quantum computing concepts, quantum circuit design, and practical implementations using quantum programming frameworks.
- Aug 2024–Sep 2025 **AI Engineer / Data Scientist**, *SoundBoost*
Designed and deployed deep learning models for audio source separation, classification, and acoustic event detection. Built end-to-end AI pipelines with Django backends for model serving and JavaScript for real-time inference. Led development of AI agents for complex audio processing workflows.
- Jul 2023–Sep 2023 **AI Engineer Intern**, *Live The World*
Engineered content generation pipelines using state-of-the-art NLP tools. Enhanced web scraping capabilities and developed Python solutions for AI-driven applications.
- Aug 2022–Jul 2023 **Data Analytics & Process Mining Intern**, *Allianz TR*
Automated Excel reporting workflows using Python and SQL. Built dynamic dashboards for operational visibility and optimized business processes using Celonis process mining.
- 2020–2022 **Research Assistant**, *Boğaziçi University*
Developed transformer-based architectures for seismic data analysis and earthquake detection. Mentored five junior researchers.
- 2021–2022 **Teaching Assistant**, *Boğaziçi University*
Led QA sessions for Numerical Methods, teaching practical applications of NumPy, SciPy, and Matplotlib through hands-on problem solving.

Education

- 2025–Present **M.Sc. in Data Science**, *Sabancı University*
Thesis research on mechanistic interpretability of large language models, studying how refusal and related concepts are represented geometrically in a model's internal activations. Coursework in advanced deep learning and statistical analysis.
- 2023–2024 **Graduate Studies in Computer Science**, *University of Padua*
Completed the first year of the M.Sc. program. Advanced coursework in Artificial Intelligence and Deep Learning, building strong theoretical foundations in deep learning architectures and algorithmic problem-solving.
- 2018–2023 **B.Sc. in Physics**, *Boğaziçi University*
Specialized in Quantum Physics and General Relativity. Completed machine learning coursework and led the Science Club.

Projects

- **TÜBİTAK 2209-A**: Developed high-precision earthquake detection model through innovative feature engineering.
- **Earth-ML**: Enhanced time series classification using advanced modeling techniques for geophysical data.
- **Kaggle ML Challenge**: Achieved 8th place in Türkiye İş Bankası ML Challenge 5 through effective feature engineering.
- **Datathon AI**: Secured 3rd place in computer vision competition.
- **NLP News Summarization**: Comparative evaluation of BART and T5 architectures for summarization tasks.
- **"Burası" Art Exhibition**: Merged seismic data with artistic representation, fusing science and art.

Skills

- **Programming:** Python, SQL, JavaScript, TypeScript
- **ML/DL:** PyTorch, XGBoost, CatBoost
- **AI/LLM:** LLM APIs (OpenAI, OpenRouter), LangChain, LangGraph, Transformers
- **Interpretability:** TransformerLens, nnsight, activation/ablation analysis
- **Backend:** FastAPI, Laravel, Django, Celery, Redis
- **Scientific:** NumPy, SciPy, Matplotlib, Pandas
- **HPC:** Slurm, multi-GPU / distributed training
- **Tools:** Git, Docker, Linux

————— Certifications

- **Quantum Computing (Bronze):** QTurkey
- **Excellence in Audio:** Hugging Face
- **Process Mining:** Celonis Academy
- **Building RAG Agents:** NVIDIA DLI

————— Languages

Turkish (Native) · English (Fluent)